

OSSP Practical-Assignment_1

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roll

1.

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[20];

    printf("Enter command: ");
    scanf("%s", cmd);

    int pid = fork();

    if (pid == 0)
    {
        printf("Child PID: %d\n", getpid());
        execlp(cmd, cmd, NULL);
    }
    else
    {
        printf("Parent PID: %d\n", getpid());
        wait(NULL);
        printf("Child completed\n");
    }

    return 0;
}
```

```
guru_kiran@LAPTOP-OTG3R078:~$ practical1.c
practical1.c: command not found
guru_kiran@LAPTOP-OTG3R078:~$ nano practical1.c
guru_kiran@LAPTOP-OTG3R078:~$ gcc practical1.c
guru_kiran@LAPTOP-OTG3R078:~$ a.out
-bash: a.out.: No such file or directory
guru_kiran@LAPTOP-OTG3R078:~$ ./a.out
Enter command: ls
Parent PID: 968
Child PID: 969
SUIIIIIIIIII.c  a8      b1  excex1.c  file12    newfile1  pipe1.c  pipe6.c  pipesort.c  public.txt  removeb2.c  z
a.c             a8.txt  b2  execp.c   fork.c    open2.c   pipe2.c  pipe7.c  practical1.c  read1.c    script.sh   z.c
a.out           allexec.c  c2  execv.c   linux_task4  orw.c    pipe3.c  pipeex.c  private.txt  readwrite.c  x2
Child completed
guru_kiran@LAPTOP-OTG3R078:~$ |
```

2.

```
guru_kiran@LAPTOP-OTG3R07B:~$ u name
u: command not found
guru_kiran@LAPTOP-OTG3R07B:~$ uname
Linux
guru_kiran@LAPTOP-OTG3R07B:~$ lscpu
Architecture:                x86_64
  CPU op-mode(s):            32-bit, 64-bit
  Address sizes:             39 bits physical, 48 bits virtual
  Byte Order:                Little Endian
CPU(s):                      12
  On-line CPU(s) list:       0-11
Vendor ID:                   GenuineIntel
Model name:                  13th Gen Intel(R) Core(TM) i5-13420H
CPU family:                  6
Model:                       186
Thread(s) per core:         2
Core(s) per socket:         6
Socket(s):                   1
Stepping:                    2
BogoMIPS:                    5222.36
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                             r pge mca cmov pat pse36 clflush mmx fxsr sse s
                             se2 ss ht syscall nx pdpe1gb rdtscp lm constant
                             _tsc rep_good nopl xtopology tsc_reliable nonst
                             op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
                             sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                             opcnt tsc_deadline_timer aes xsave avx f16c rdr
                             and hypervisor lahf_lm abm 3dnowprefetch ssbd i
                             brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
                             d ept_ad fsgsbase tsc_adjust bml avx2 smep bmi
                             2 erms invpcid rdseed adx smap clflushopt clwb
                             sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
                             vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo
                             vdiri movdir64b fsrm md_clear serialize ibt flu
                             sh_lld arch_capabilities

Virtualization features:
  Virtualization:            VT-x
  Hypervisor vendor:         Microsoft
  Virtualization type:       full
Caches (sum of all):
  L1d:                       288 KiB (6 instances)
  L1i:                       192 KiB (6 instances)
  L2:                         7.5 MiB (6 instances)
  L3:                        12 MiB (1 instance)
NUMA:
  NUMA node(s):              1
  NUMA node0 CPU(s):         0-11
```

```

NUMA:
  NUMA node(s):          1
  NUMA node0 CPU(s):     0-11
Vulnerabilities:
  Gather data sampling:   Not affected
  Ghostwrite:            Not affected
  Indirect target selection: Not affected
  Itlb multihit:         Not affected
  L1tf:                  Not affected
  Mds:                   Not affected
  Meltdown:              Not affected
  Mmio stale data:       Not affected
  Old microcode:         Not affected
  Reg file data sampling: Mitigation; Clear Register File
  Retbleed:               Mitigation; Enhanced IBRS
  Spec rstack overflow:   Not affected
  Spec store bypass:     Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:            Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:            Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
  Srbds:                 Not affected
  Tsa:                   Not affected
  Tsx async abort:       Not affected
  Vmscape:               Not affected
guru_kiran@LAPTOP-OTG3R07B:~$ lsblk
NAME MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda   8:0      0 356.9M  1 disk
sdb   8:16     0 159.4M  1 disk
sdc   8:32     0    2G  0 disk [SWAP]
sdd   8:48     0    1T  0 disk /mnt/wslg/distro
/

```

```
guru_kiran@LAPTOP-0TG3R07B:~$ top
top - 06:03:13 up 40 min, 1 user, load average: 0.08, 0.05, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.1 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7754.6 total, 7145.2 free, 552.6 used, 207.8 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7202.0 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24208   15292   11536 S   0.0   0.2   0:00.57 systemd
    2 root        20   0    3180    2200    2072 S   0.0   0.0   0:00.01 init-systemd(Ub
    6 root        20   0    3180    2044    1940 S   0.0   0.0   0:00.00 init
   43 root        19  -1   50392   16820   15540 S   0.0   0.2   0:00.19 systemd-journal
   78 systemd+    20   0   22540   14652   12032 S   0.0   0.2   0:00.08 systemd-resolve
   86 root        20   0   35232   12312    9232 S   0.0   0.2   0:00.78 systemd-udev
  126 root        20   0    2888    1944    1820 S   0.0   0.0   0:00.03 chronyd-starter
  133 root        20   0    4472    3164    2916 S   0.0   0.0   0:00.01 cron
  141 message+   20   0    8816    5396    4724 S   0.0   0.1   0:00.04 dbus-daemon
  154 root        20   0   44092   29780   14644 S   0.0   0.4   0:00.21 networkd-dispat
  168 root        20   0   18432    9384    8140 S   0.0   0.1   0:00.07 systemd-logind
  189 syslog      20   0  220548    5648    4352 S   0.0   0.1   0:00.07 rsyslogd
  240 _chrony      20   0   20420    6324    5548 S   0.0   0.1   0:00.07 chronyd
  253 root        20   0  123456   32484   17872 S   0.0   0.4   0:00.15 unattended-upgr
  267 root        20   0    5492    2764    2580 S   0.0   0.0   0:00.00 agetty
  268 _chrony      20   0   12092    2456    1812 S   0.0   0.0   0:00.00 chronyd
  317 root        20   0    3184    1104     980 S   0.0   0.0   0:00.00 SessionLeader
  318 root        20   0    3200    1244    1108 S   0.0   0.0   0:00.06 Relay(319)
  319 guru_ki+    20   0    6268    5604    3928 S   0.0   0.1   0:00.03 bash
  320 root        20   0    8648    5536    4692 S   0.0   0.1   0:00.01 login
  365 guru_ki+    20   0   22504   13424   10980 S   0.0   0.2   0:00.10 systemd
  367 guru_ki+    20   0   22908    4076    2076 S   0.0   0.1   0:00.00 (sd-pam)
  396 guru_ki+    20   0    6136    5428    3864 S   0.0   0.1   0:00.01 bash
  551 guru_ki+    20   0    8328    6132    3904 R   0.0   0.1   0:01.01 top
  766 root        20   0 1792044   15260   12556 S   0.0   0.2   0:00.06 wsl-pro-service
```

Linux OS acts as a link between computer hardware and applications.

Linux manages the CPU using process scheduling.

Linux manages storage using files, directories, and file systems.

uname is used to check the Linux kernel and system information.

lscpu gives details about the CPU, cores, threads, and architecture.

lsblk shows the storage devices, disks, SSDs, and partitions.

ps is used to see the running processes and their PIDs.

top shows the real-time CPU, memory, and process usage.